Technische Hochschule Augsburg

STATISTICS

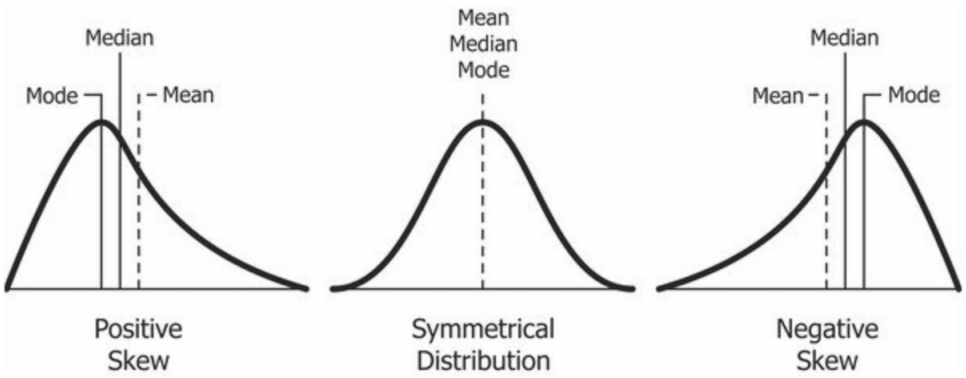
International Information Systems, 3rd semester

(6) Elements of Probability Theory

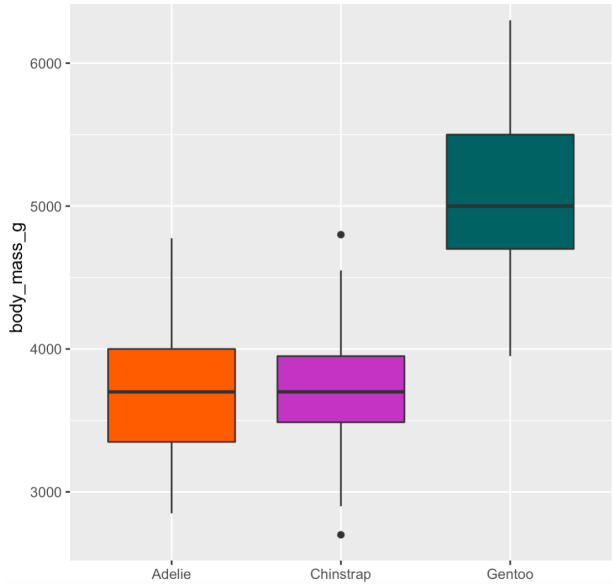
PROF. DR. -PHIL ALESSANDRA ZARCONE

LAST WEEK

Descriptive statistics



- The purpose of **descriptive statistics** is to summarize and visualize the data and understand patterns
  - **plot the frequency of your data (continuous)** to show the **shape** of your data or use a **frequency table (categorical)**
  - show the central tendency with **mean**, **median** and **mode**
  - look at the **spread or dispersion**
  - locate and possibly remove **outliers**



Questions?

Homework solutions in
R_exercise_6_solution.R

PRELIMINARY COURSE PLAN

- Introduction, Descriptive statistics: measures of central tendency
- Introduction to R, Descriptive statistics: measures of dispersion, distributions, visualizations
- **[Oct 23rd] Basic probability theory**
- [Oct 30th] Introduction to inferential statistics: sampling, hypothesis testing
- [Nov 6th] Test Statistics (I)
- [Nov 13th] **No class *(Tag der Lehre)***
- [Nov 20th-Nov 27th] Test statistics (II-III)
- [Dec 4th] Introduction to modeling: correlation
- [Dec 11th-18th] The Generalized Linear Model
- [Jan 9th-29th] Extra session if needed, Mock Exam, QA, extra exercises

SAMPLE AND POPULATION

“In this quarter, the purchase of vegan candy has increased compared to last year”

=

population *sample*

“In Germany, the purchase of vegan candy has increased compared to last year”

>

population *sample*

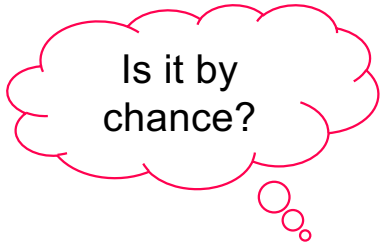
Inferential Statistics

WHY PROBABILITY THEORY?

- When we get to inferential statistics, we will rely heavily on **statements of probability**

- Let's say we observe **something in the sample** (e.g. *one candidate winning the elections*, *more vegan than non-vegan candy*): **can we say this is also true *of the population*?**
  - Let's say we found out candidate Smith will win the election according to **the data of the sample**
  - We want to know if this is also reflected in **the data of the population**
  - We look for **contrary evidence**: what's the **probability** it that this result was obtained **by chance alone** (and that the population ***does not actually*** follow this pattern)?

- This is why we need to **understand probabilities** and a few basic rules for **computing and manipulating them**



TODAY'S TOPICS AND LEARNING GOALS

Three views of probability

- Understand and explain the ***analytic view***, the ***frequentist view*** and the ***Bayesian view*** of probability

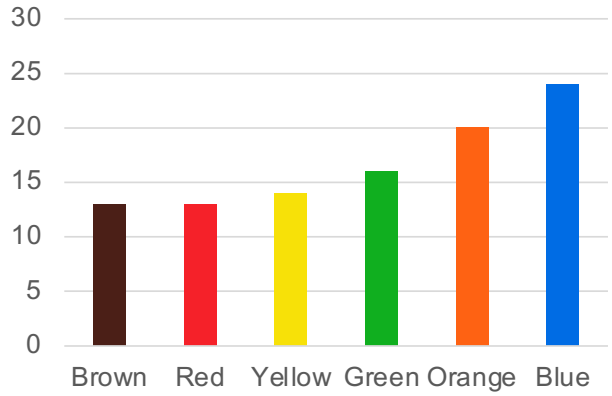
Independent events

- Computing and manipulating probabilities of independent events

Conditional probabilities

- Understand and compute the ***accuracy***, ***sensitivity***, ***specificity*** and ***precision*** of a test
- Explain how believes are updated with the ***Bayes rule***

THREE VIEWS OF PROBABILITY

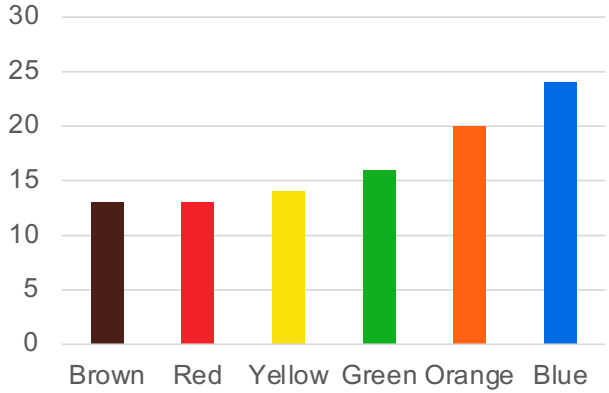


AN ANALYTIC VIEW OF PROBABILITY

One way to look at probability is as an **analysis of possible outcomes**:

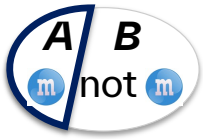
- Suppose that you have a bag of 100 M&M's and you pull one out randomly
- If you know how many M&Ms are in the bag,
  what is the probability that you will pull out **a blue M&M** ?
  - **24%** of the M&M's are **blue**
  - probability values range
    from 0 (there's ***no chance*** of the event happening)
    to 1 (the event will ***definitely*** happen)
  - the probability of drawing **a blue M&M** is **0.24**

Brown	Red	Yellow	Green	Orange	Blue
13	13	14	16	20	24
13%	13%	14%	16%	20%	24%



AN ANALYTIC VIEW OF PROBABILITY

The analytic view is based on **how many ways there are** to pull **a blue M&M** (***A*** ways) and how many ways of pulling **a non-blue M&M** (***B*** ways)



- The probability of any M&M (m or not m) is **1**
- The probability of m is **0.24**
- The probability of not m is 1 – 0.24 = **0.76**

$$P(\text{m or not m}) = A + B$$

$$P(\text{m}) = \frac{A}{A+B}$$

$$P(\text{not m}) = 1 - P(\text{m})$$

$$P(\text{not m}) = \frac{B}{A+B}$$

Brown	Red	Yellow	Green	Orange	Blue
0.13	0.13	0.14	0.16	0.20	0.24

A FREQUENTIST VIEW OF PROBABILITY

- **Sample with replacement**: we start randomly picking a M&M from the bag, taking note of the color and putting back in the bag
- If we made ***a very large number of draws***, we would find that 24% of them is blue
- The frequentist view defines probability in terms of **past performance**
  - it is the ***limit of the relative frequency of occurrence*** of an event, which we ***approach*** as the number of draws increases

Brown	Red	Yellow	Green	Orange	Blue
𝍸 𝍸 𝍸	𝍸 𝍸 𝍸	𝍸 𝍸 𝍸	𝍸 𝍸 𝍸	𝍸 𝍸 𝍸	𝍸 𝍸 𝍸
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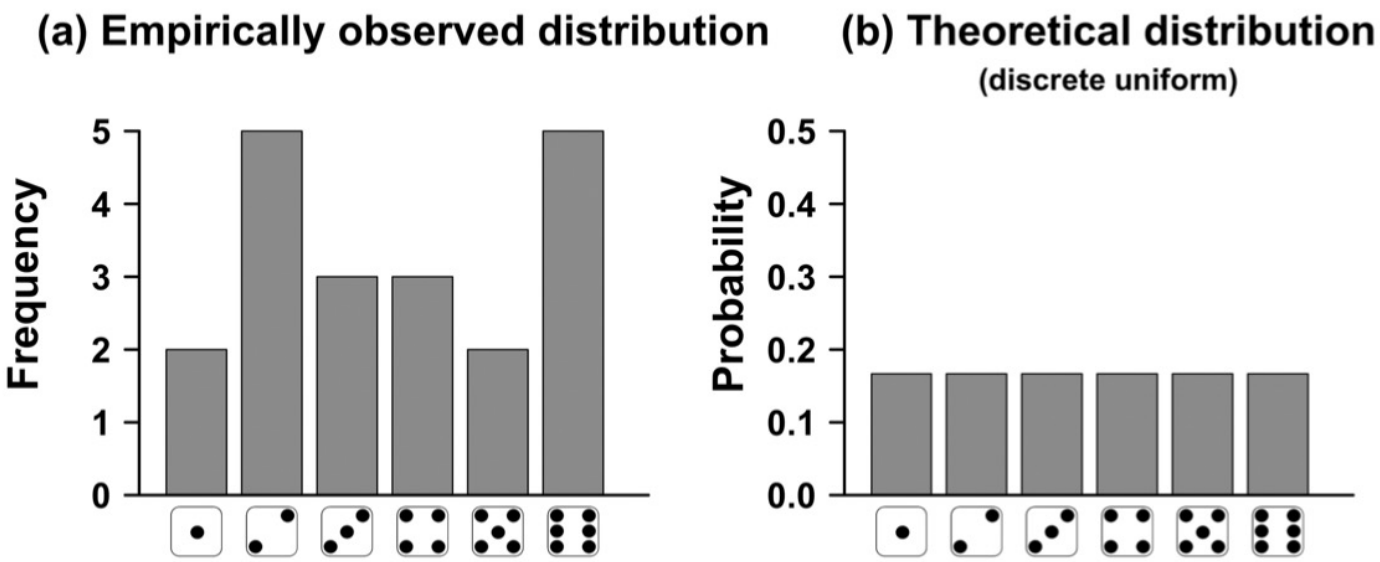
B. Winter, Statistics for Linguists

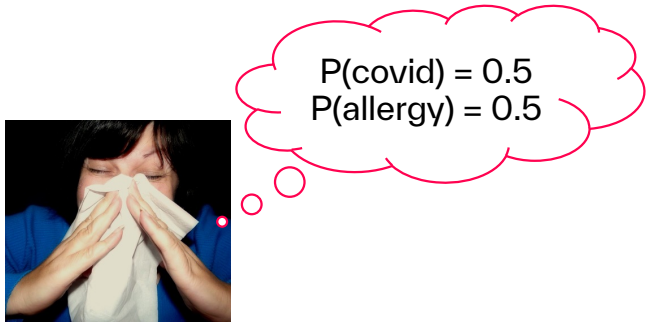
A FREQUENTIST VIEW OF PROBABILITY

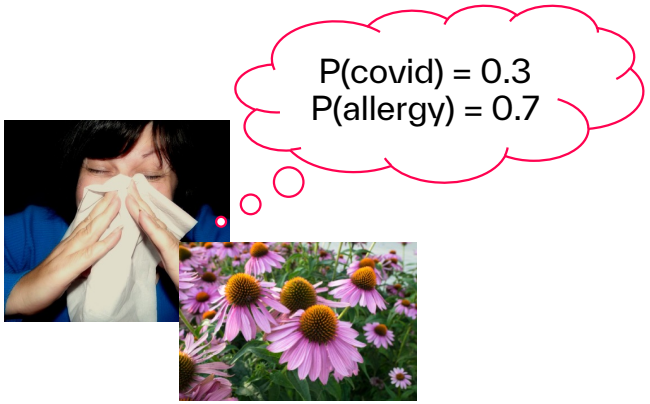


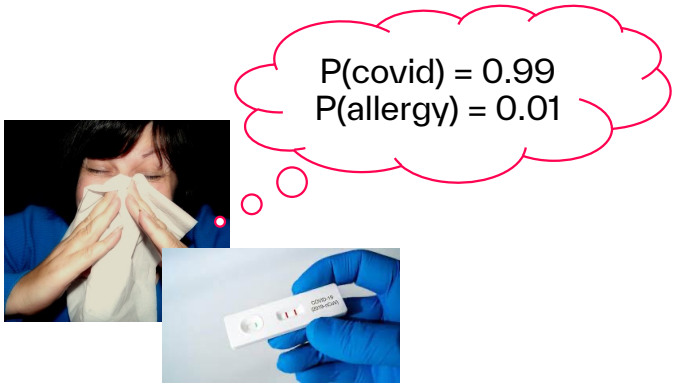
With a high number of empirically-observed events, we approach the theoretical distribution

A SUBJECTIVE VIEW OF PROBABILITY (BAYESIAN VIEW)

- Not everything is as simple as tossing a coin or casting a die
  - sometimes we have to ***estimate the likelihood*** of a future event based on **evidence** (prior knowledge, subjective)
  - Subjective probabilities play an extremely important role in **human decision making**







- The subjective view of probability defines probability in terms of ***personal subjective belief*** in the likelihood of an outcome

Questions?

INDEPENDENT EVENTS: COMPUTING AND MANIPULATING PROBABILITIES

TERMINOLOGY

- Probability deals with **events**
  - pulling out a blue M&M
  - rolling 6 on a die
  - getting an ace out of a deck of cards
  - passing an exam

- Events are **mutually exclusive** if the occurrence of one precludes the occurrence of the other (e.g. rolling 6 and rolling 1 on the same die simultaneously)
- A set of mutually exclusive events (**exhaustive set**) includes **all possible outcomes**
- Events are **independent** if the occurrence (or non-occurrence) of one has no effect on the occurrence (or non-occurrence) of the other

THE ADDITIVE LAW OF PROBABILITY

Brown	Red	Yellow	Green	Orange	Blue
0.13	0.13	0.14	0.16	0.20	0.24

↗ What if I am ok with either a or a ?

Additive law of probability
Given a set of ***mutually exclusive events***,
the probability of occurrence of one ***or*** another in the set
is equal to the **sum** of their separate probabilities

$$P(\text{m} \text{ or } \text{m}) = P(\text{m}) + P(\text{m})$$

$$P(\text{m} \text{ or } \text{m}) = 0.13 + 0.24 = 0.37$$

THE MULTIPLICATIVE LAW OF PROBABILITY

Brown	Red	Yellow	Green	Orange	Blue
0.13	0.13	0.14	0.16	0.20	0.24

- What if I pick two M&M's (sampling with replacement)

  and I want the first to be a and the second to be a ?

Multiplicative law of probability
Given a set of ***independent events***,
their **joint probability** (probability of their joint occurrence)
is equal to the **product** of their separate probabilities

$$P(\text{m}, \text{m}) = P(\text{m}) \times P(\text{m})$$

$$P(\text{m}, \text{m}) = 0.13 \times 0.24 = 0.0312$$

- What's the probability that both the first and the second draw will be ?

$$P(\text{m}, \text{m}) = 0.13 \times 0.13 = 0.0169$$

USING THE ADDITIVE AND THE MULTIPLICATIVE LAW OF PROBABILITY

Brown	Red	Yellow	Green	Orange	Blue
0.13	0.13	0.14	0.16	0.20	0.24

- What if I pick two M&M's (sampling with replacement)

  and I want the first to be a and the second to be a ,

  but the order doesn't matter?

$P($ $,$ $) = P($ $) \times P($ $) = 0.13 \times 0.24 = 0.0312$

$P($ $,$ $) = P($ $) \times P($ $) = 0.24 \times 0.13 = 0.0312$

$P($ $,$ $) + P($ $,$ $) = 0.0312 + 0.0312 = 0.0624$

EXERCISE

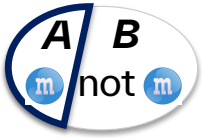
(consider a year to be made of 365 days - ignore February 29th - and assume sampling with replacement)

1. You want to ask the person next to you when their birthday is. Before you ask, what's the probability...
   - ... that today is their birthday?
   - ... that today is *not* their birthday?
2. If you now know that there are 100 people in IIS, and 15 are born in spring, 30 in summer, 35 in fall and 20 in winter, what's the probability...
   - ... that a random IIS student is born in fall *or* winter?
   - ... that of two random IIS students one is born in fall and the other in winter (order doesn't matter)?

Questions?

CONDITIONAL PROBABILITIES: SENSITIVITY AND SPECIFICITY OF A TEST

INDEPENDENT EVENTS VS. NON INDEPENDENT EVENTS



- Events are ***independent*** if the occurrence (or non-occurrence) of one has no effect on the occurrence (or non-occurrence) of the other

- If two events are ***not independent***, we talk of **Conditional Probability**: the probability of a particular event happening if another event has also happened, e.g.
  - probability of rain given overcast weather
  - probability of having covid given a positive test
  - probability of a m after a m if we do not put the first one back (sampling ***without*** replacement)

Probability of B given A

$$P(B|A) = \frac{P\,(A \cap B)}{P(A)}$$

Probability of A and B happening together

Probability of A

NON INDEPENDENT EVENTS

- Probability of rain given overcast weather

$$P(\text{rain}|\text{clouds}) = \frac{P\ (\text{cloudy} \cap \text{rain})}{P(\text{cloudy})} \qquad P(\text{rain}|\text{sunny}) = \frac{P\ (\text{sunny} \cap \text{rain})}{P(\text{sunny})}$$

- Probability of rain given overcast weather

$$P(\text{sneezing}|\text{pollen}) = \frac{P\ (\text{pollen} \cap \text{sneezing})}{P(\text{pollen})} \qquad P(\text{sneezing}|\text{winter}) = \frac{P\ (\text{winter} \cap \text{sneezing})}{P(\text{winter})}$$

- Probability of the results in the sample, if this was due to chance

$$P(\text{sneezing}|\text{chance})$$

SAMPLING WITH AND WITHOUT REPLACEMENT

Brown	Red	Yellow	Green	Orange	Blue
0.13	0.13	0.14	0.16	0.20	0.24

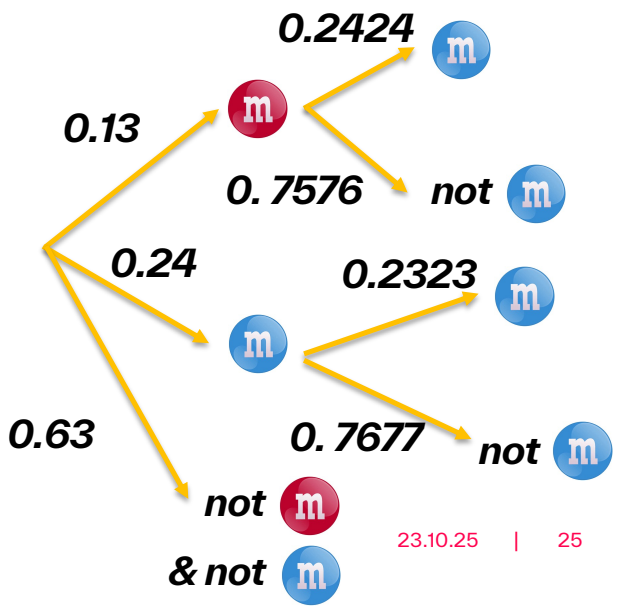
- If I pick two M&M's (sampling **with** replacement) and I want the first to be a m and the second to be a m ?
  - Joint probability of **independent** events:

$$P(m, m) = P(m) \times P(m) =$$
$$= 0.13 \times 0.24 = 0.0312$$

- If I pick two M&M's (sampling **without** replacement) and I want the first to be a m and the second to be a m ?
  - **Conditional** probability: the probability of m after removing a m is different than before!

$$P(m, m) = P(m) \times P(m \mid m) = P(m) \frac{P(m \cap m)}{P(m)}$$

$$= 0.13 \times \frac{0.0315}{0.13} = 0.13 \times 0.2424 = 0.03151$$



https://www.cochrane.org/CD013705/INFECTN_how-accurate-are-rapid-antigen-tests-diagnosing-covid-19

SENSITIVITY AND SPECIFICITY OF A TEST

2x2 Confusion Matrix

1000 people with symptoms who had an antigen test

		Predicted class	
		Positive test	Negative test
Actual class	Infected	40	10
	Not infected	5	945

		Predicted class	
		Positive test	Negative test
Actual class	Infected	True Positives (TP)	False Negatives (FN)
	Not infected	False Positives (FP)	True Negatives (TN)

https://www.cochrane.org/CD013705/INFECTN_how-accurate-are-rapid-antigen-tests-diagnosing-covid-19

SENSITIVITY AND SPECIFICITY OF A TEST

2x2 Confusion Matrix

Accuracy: $\frac{\text{correct}}{\text{all}}$

1000 people with symptoms who had an antigen test

		Predicted class	
		Positive test	Negative test
Actual class	Infected	40	10
	Not infected	5	945

aka Recall or True Positive Rate

Sensitivity

$P(\text{pos_test}|\text{infected})$

$$P(\text{pos_test}|\text{infected}) = \frac{\text{pos_test} \cap \text{infected}}{\text{infected}} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

How many positive cases were we able to recall / retrieve?

https://www.cochrane.org/CD013705/INFECTN_how-accurate-are-rapid-antigen-tests-diagnosing-covid-19

SENSITIVITY AND SPECIFICITY OF A TEST

2x2 Confusion Matrix

1000 people with symptoms who had an antigen test

		Predicted class	
		Positive test	Negative test
Actual class	Infected	40	10
	Not infected	5	945

Accuracy: $\frac{\text{correct}}{\text{all}}$

Precision

$P(\text{infected}|\text{pos_test})$

aka Positive Predictive Value

Sensitivity

$P(\text{pos_test}|\text{infected})$

How many detected cases were correct?

$$P(\text{infected}|\text{pos_test}) = \frac{\text{infected} \cap \text{pos_test}}{\text{pos_test}} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

https://www.cochrane.org/CD013705/INFECTN_how-accurate-are-rapid-antigen-tests-diagnosing-covid-19

SENSITIVITY AND SPECIFICITY OF A TEST

2x2 Confusion Matrix

Accuracy: $\frac{\text{correct}}{\text{all}}$

1000 people with symptoms who had an antigen test

		Predicted class	
		Positive test	Negative test
Actual class	Infected	40	10
	Not infected	5	945

Precision
$P(\text{infected}|\text{pos_test})$

Sensitivity
$P(\text{pos_test}|\text{infected})$

Specificity
$P(\text{neg_test}|\text{not_infected})$

aka True Negative Rate

$$P(\text{neg_test}|\text{not_infected}) = \frac{\text{neg_test} \cap \text{not_infected}}{\text{not_infected}} = \frac{\text{TN}}{\text{FP} + \text{TN}}$$

How many cases declared negative were correct?

https://www.cochrane.org/CD013705/INFECTN_how-accurate-are-rapid-antigen-tests-diagnosing-covid-19

SENSITIVITY AND SPECIFICITY OF A TEST

2x2 Confusion Matrix

Accuracy: $\frac{\text{correct}}{\text{all}}$

1000 people with symptoms who had an antigen test

		Predicted class	
		Positive test	Negative test
Actual class	Infected	40	10
	Not infected	5	945

Precision
$P(\text{infected}|\text{pos_test})$

Sensitivity
$P(\text{pos_test}|\text{infected})$

Specificity
$P(\text{neg_test}|\text{not_infected})$

Condition: the occurrence of one event influences another

$$P(\text{pos_test}|\text{infected}) = \frac{\text{pos_test} \cap \text{infected}}{\text{infected}} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$P(\text{neg_test}|\text{not_infected}) = \frac{\text{neg_test} \cap \text{not_infected}}{\text{not_infected}} = \frac{\text{TN}}{\text{FP} + \text{TN}}$$

$$P(\text{infected}|\text{pos_test}) = \frac{\text{infected} \cap \text{pos_test}}{\text{pos_test}} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Evidence

EXERCISE: ASYMPTOMATIC CASES

		Predicted class	
		Positive test	Negative test
Actual class	Infected	32	18
	Not infected	30	9920

- What's the accuracy?
- What's the specificity?
- What's the sensitivity?
- What's the precision?

PRECISION AND RECALL OF A MULTICLASS CLASSIFIER

		Predicted class		
		🐶	🐱	🐴
Actual class	🐶	8	5	3
	🐱	10	60	30
	🐴	1	50	200

↗ **Accuracy:** $\frac{\text{correctly_labeled}}{\#\,\text{all}} = \frac{268}{367} = 0.73$

↗ **Precision:** $P(🐶) = \frac{8}{19} = 0.42$ $\quad P(🐱) = \frac{60}{115} = 0.52$ $\quad P(🐴) = \frac{200}{233} = 0.86$

↗ **Recall:** $R(🐶) = \frac{8}{16} = 0.5$ $\quad R(🐱) = \frac{60}{100} = 0.6$ $\quad R(🐴) = \frac{200}{251} = 0.797$

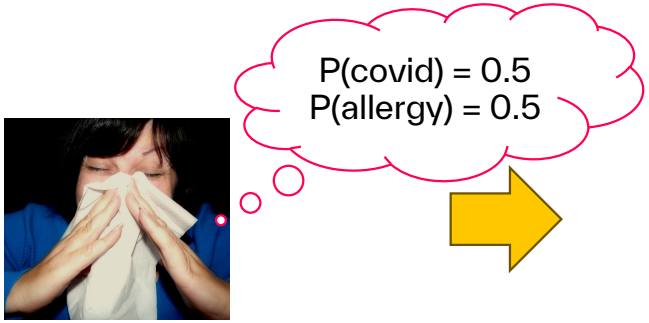
Questions?

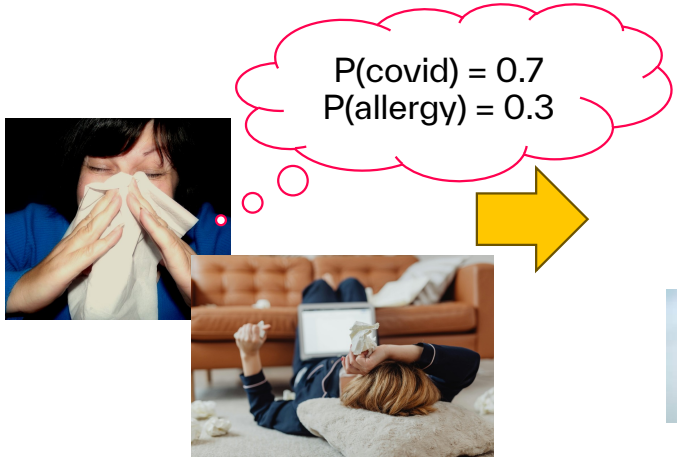
CONDITIONAL PROBABILITIES: BAYESIAN STATISTICS

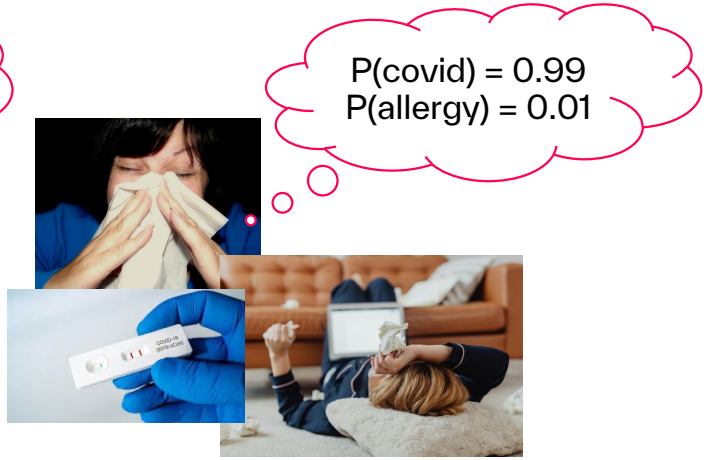
UPDATING YOUR BELIEFS WITH BAYES

Thomas Bayes

- Subjective view of statistics:
  - We all have ***personal subjective beliefs*** in the likelihood of an outcome
  - But our beliefs ***can change*** when we are exposed to ***experience (observing evidence)***
  - The **Bayesian approach** offers us tools to deal with this







UPDATING YOUR BELIEFS WITH BAYES

Thomas Bayes

- Subjective view of statistics:
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Bayes' Theorem

Likelihood
P of the evidence given the hypothesis

a-priori
P of the hypothesis before the observation

a-posteriori
P of the hypothesis given the evidence

$$P(h|e) = \frac{P(e|h)P(h)}{P(e)}$$

P of the evidence

Strength of the evidence

$$P(\text{infected}|\text{pos_test}) = \frac{P(\text{pos_test}|\text{infected})P(\text{infected})}{P(\text{pos_test})} = \frac{0.8 \times 0.05}{0.045} = 0.89$$

WHY DOES THIS WORK?

$$P(A|B) = \frac{P\,(B \cap A)}{P(B)}$$

$$P(B|A) = \frac{P\,(A \cap B)}{P(A)}$$

$$P(B|A)P(A) = \frac{P\,(A \cap B)}{\cancel{P(A)}}\cancel{P(A)}$$

$$P\,(B \cap A) = P\,(A \cap B)$$

$$P\,(A \cap B) = P(B|A)P(A)$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

If P(B|A) is known but P(A|B) is not, this is helpful because it "turns" P(A|B) into P(B|A)

SUMMING UP (1)

- We can think of probability adopting an **analytical view**, a **frequentist view**, or a **subjective view**
- **Independent events**
  - The sum of the probabilities of a set of ***mutually exclusive*** events sums up to 1

$$P(\text{m}) + P(\text{not m}) = 1$$

$$P(\text{m}) = 1 - P(\text{not m})$$

Additive law of probability
Given a set of ***mutually exclusive events***,
the probability of occurrence of one ***or*** another in the set
is equal to the **sum** of their separate probabilities

$$P(\text{m or m}) = P(\text{m}) + P(\text{m})$$

Multiplicative law of probability
Given a set of ***independent events***,
their **joint probability** (probability of their joint occurrence)
is equal to the **product** of their separate probabilities

$$P(\text{m} \cap \text{m}) = P(\text{m}) \times P(\text{m})$$

SUMMING UP (2)

- **Non-independent events**
  - If two events are ***not independent***, we talk of **Conditional Probability**:
    the probability of a particular event happening if another event has also happened

$$P(B|A) = \frac{P\ (A \cap B)}{P(A)}$$

  - Using conditional probabilities we can compute the
    sensitivity $P(\text{pos_test}|\text{pos_result})$,
    specificity $P(\text{neg_test}|\text{neg_result})$ and **precision** $P(\text{pos_result}|\text{pos_test})$ of a test

  - Using **Bayes' Theorem**, we can compute the
    conditional probability of $P(\text{A}|\text{B})$ knowing $P(\text{B}|\text{A})$

Questions?

Exercise:
R_exercise_7.R
BreastCancerTest.csv

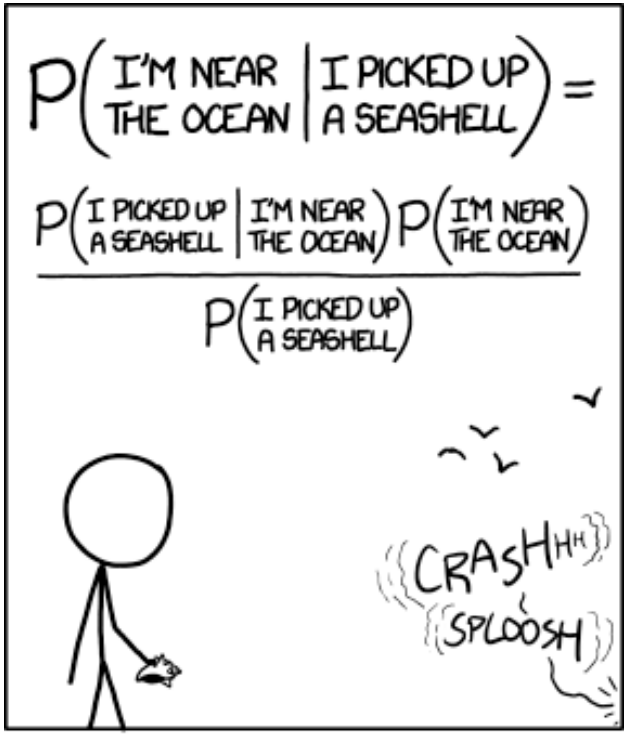
HOMEWORK

- Chapter 7, sections 7.1 to 7.3 of Howell, “Fundamental Statistics for the Behavioral Sciences” (available on Moodle)

- Watch this videos:
  - “Mathematical Thinking” (Crash Course Statistics) https://www.youtube.com/watch?v=tN9Xl1AcSv8
  - “Probability Part 1: Rules and Patterns” (Crash Course Statistics) https://www.youtube.com/watch?v=OyddY7DIV58
  - “Probability Part 2: Updating Your Beliefs with Bayes” (Crash Course Statistics) https://www.youtube.com/watch?v=oZCskBpHWyk

- Exercise in slides 20 and 31 (solutions in slides 42 and 43)

Source: xkcd



STATISTICALLY SPEAKING, IF YOU PICK UP A SEASHELL AND *DON'T* HOLD IT TO YOUR EAR, YOU CAN PROBABLY HEAR THE OCEAN.

EXERCISE

(consider a year to be made of 365 days - ignore February 29th - and assume sampling with replacement)

1. You want to ask the person next to you when their birthday is. Before you ask, what's the probability...
   - ... that today is their birthday? $\frac{1}{365} = 0.0027$
   - ... that today is *not* their birthday? $1 - \frac{1}{365} = 1 - 0.0027 = 0.9973$

2. If you now know that there are 100 people in IIS, and 15 are born in spring, 30 in summer, 35 in fall and 20 in winter, what's the probability...
   - ... that a random IIS student is born in fall *or* winter? $P(\text{fall_or_winter}) = P(\text{fall}) + P(\text{winter}) = \frac{35}{100} + \frac{20}{100} = 0.55$
   - ... that of two random IIS students one is born in fall and the other in winter (order doesn't matter)?

$$P(\text{fall,winter}) + P(\text{winter,fall}) = P(\text{fall}) * P(\text{winter}) + P(\text{winter}) * P(\text{fall}) = 0.35 * 0.2 + 0.2 * 0.35 = 0.14$$

EXERCISE: ASYMPTOMATIC CASES

		Predicted class	
		Positive test	Negative test
Actual class	Infected	32	18
	Not infected	30	9920

- What's the accuracy?

$$\text{Accuracy} = \frac{9952}{10000} = 0.9952$$

- What's the sensitivity?

$$P(\text{pos_test}|\text{infected}) = \frac{\text{pos_test} \cap \text{infected}}{\text{infected}}$$
$$= \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{32}{50} = 0.64$$

- What's the specificity?

$$P(\text{neg_test}|\text{not_infected}) = \frac{\text{neg_test} \cap \text{not_infected}}{\text{not_infected}} =$$
$$= \frac{\text{TN}}{\text{FP} + \text{TN}} = \frac{9920}{9950} = 0.997$$

- What's the precision?

$$P(\text{infected}|\text{pos_test}) = \frac{\text{infected} \cap \text{pos_test}}{\text{pos_test}}$$
$$= \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{32}{62} = 0.516$$